Vignesh Ram Sivakumar

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SUMMARY

MSc Data Science student at King's College London (71.7% Sem 1, Distinction trajectory) building DAMS-GCT: an uncertainty-aware GNN-Transformer for multimodal emotion recognition trained on NVIDIA A100 GPUs. Work spans vision, speech, and physiological signal AI, with hands-on PyTorch, uncertainty quantification, and end-to-end ML from research prototype to containerised REST API deployment. Available for full-time AI/ML Engineer roles in London from August 2026.

EDUCATION

MSc in Data Science — King's College London *Sep 2025 – Sep 2026*

Expected: Distinction · Sem 1 average: 71.7% (Distinction in 3 subjects: Computer Programming for Data Scientists, Introduction to Data Visualisation, Telling Stories with Data)

Key Modules: Neural Networks & Deep Learning · Data Mining · Big Data Technologies · Databases & Data Warehousing · Statistics

Dissertation: Uncertainty-Aware Multimodal Emotion Recognition using GNNs & Physiological Signals (Supervisor: Dr. Helen Yannakoudakis)

B.Tech in Computer Science and Engineering *Jun 2021 – May 2025*

SRM Institute of Science & Technology · Chennai, India

GPA: 8.73/10 — First Class with Distinction

EXPERIENCE

ML Research Engineer (MSc Dissertation) *Jan 2026 – Present*

King's College London · London, UK

- Designed DAMS-GCT (Dynamic Adaptive Multi-Scale Graph-Convolutional Transformer) for 7-class emotion recognition, fusing EEG (310-dim DE features), EDA, and BVP signals from the SEED-VII dataset across 20 subjects — trained on KCL's HPC cluster using NVIDIA A100 GPUs.
- Implemented Monte Carlo Dropout for uncertainty quantification across spatio-temporal GNN layers, capturing EEG inter-electrode spatial dependencies and producing calibrated confidence scores per emotion prediction.
- Improved generalisation to 20+ unseen subjects using cross-dataset transfer learning and modality-dropout augmentation, with subject-independent benchmarks validated on SEED-VII.
- Benchmarked against the published MAET baseline (52.46% on SEED-VII), measuring how normalisation strategy and 4-fold cross-validation protocol each affect subject-dependent results.

Technologies: Python, PyTorch, GNNs, EEG/EDA/BVP Signal Processing, Uncertainty Quantification, Monte Carlo Dropout, Transfer Learning, Weights & Biases, SLURM/HPC

AI Intern — Clustering and Deep Learning Analysis *Dec 2023 – Jan 2024*

4i Apps Solutions Private Limited · Chennai, India · Hybrid

- Built a K-Means segmentation pipeline on ∼3–4 GB of purchase data, identifying customer behavioural patterns and translating findings into targeting recommendations for the business team.
- Trained and evaluated TensorFlow/Keras deep learning models for image classification and NLP tasks, taking each from raw data through preprocessing, training, and final performance reporting.

Data Science Intern — Data Science & Machine Learning *Jul 2023*

4i Apps Solutions Private Limited · Chennai, India · Hybrid

- Built an end-to-end classification pipeline on commercial datasets: data cleaning and EDA with Pandas/NumPy, then model development and train/test evaluation with Scikit-Learn.
- Built dashboards in Matplotlib and Seaborn to present model results to clients, producing the charts and summaries used in business reporting.

PROJECTS

DAMS-GCT: Uncertainty-Aware Multimodal Emotion Recognition [MSc Dissertation] *Jan 2026 – Present*

King's College London · Supervisor: Dr. Helen Yannakoudakis

- Developed the DAMS-GCT architecture: cross-modal attention fusion over spatio-temporal GNNs and Transformer encoders, processing EEG (310-dim) and eye-movement signals (33-dim) jointly on SEED-VII (20 subjects, 7 emotion classes).
- Reproduced the MAET model end-to-end to verify the published 52.46% accuracy baseline, measuring how intensity filtering, normalisation, and cross-validation protocol affect subject-dependent results.

Technologies: Python, PyTorch, GNNs, Transformers, EEG Signal Processing, Uncertainty Quantification, Transfer Learning, Weights & Biases

Conversational Multimodal Image Recognition Chatbot *Jan 2025 – May 2025*

SRM Institute of Science and Technology · Supervisor: Dr. J. Selvin Paul Peter · github.com/V1cky16112003/Conversational-Image-Recognition-Chatbot

- Built a multimodal chatbot combining real-time image recognition (Gemini 2.0 Flash), OCR (Azure Computer Vision), and voice I/O (Azure Speech) with support for English, Tamil, Hindi, and Telugu.
- Designed a production REST API (FastAPI backend, React.js frontend) that handles real-time multimodal inference across 4 languages with sub-second response latency.

Technologies: Gemini 2.0 Flash, Azure Cognitive Services, FastAPI, React.js, Python, OCR, NLP, TTS, STT

Automated Lecture Transcription & Summarisation System *Aug 2024 – Nov 2024*

SRM Institute of Science and Technology · Supervisor: Dr. J. Selvin Paul Peter · github.com/V1cky16112003/Automated-Lecture-Transcription-and-Summarization-Sytsem-with-Multilingual-Support

- Built an automated pipeline using OpenAI Whisper (transcription), T5 (summarisation), and Google Translate API (translation) across 4 languages (English, Tamil, Hindi, Telugu). Tested on live TedX audio, reaching 95% transcription accuracy.

Technologies: OpenAI Whisper, T5 Summarisation, Google Translate API, Python, NLP, Neural Machine Translation

TECHNICAL SKILLS

Programming: Python (proficient), SQL, JavaScript, C/C++, Java, Git

ML & AI: PyTorch, TensorFlow, Scikit-Learn, GNNs, Transformers, HuggingFace, Multimodal AI, NLP, Computer Vision, Uncertainty Quantification

Signals & Modelling: EEG/EDA/BVP signal processing, Multimodal fusion, Uncertainty quantification, Monte Carlo Dropout, Transfer learning

Engineering & Backend: FastAPI, Docker, REST APIs, Oracle Cloud, Azure Cognitive Services, MySQL

Frontend & Viz: React.js, D3.js, Tableau, Matplotlib, Seaborn

CERTIFICATIONS

Oracle Cloud Infrastructure 2023 Certified Foundations Associate · Oracle *Apr 2024*

AWS Academy Machine Learning Foundations · Amazon Web Services *Apr 2023*